# Few-Shot Remote Sensing Scene Classification With Spatial Affinity Attention and Class Surrogate-Based Supervised Contrastive Learning

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***Abstract*— Learning powerful and discriminative representation is critical for boosting the performance of few-shot remote sensing scene classification (FSRSSC). The remote sensing images have unique characteristics, such as, complex background and co-occurrence of multiple objects, making FSRSSC challenging. To address this problem, in this article, a novel FSRSSC method is proposed. First, a spatial affinity attention (SAA) mechanism is designed to encourage the network model to focus on critical regions. The SAA infers attention maps from both channel and spatial dimensions, and encodes the mean values and affinities of feature nodes in each channel along the vertical and horizontal directions. Second, a class surrogate-based supervised contrastive learning (CSSCL) strategy is proposed to promote intra-class compactness and inter-class dispersion. Different from supervised contrastive learning (SCL), the CSSCL learns a surrogate for each class in the contrastive space and select the corresponding class surrogate rather than the samples from the same class for the anchor to form positive pairs. It can alleviate the impact of too hard or too simple positive sample pairs on the model generalization that exists when SCL is introduced into FSRSSC. The model is trained under the joint supervision of cross-entropy (CE) loss and CSSCL loss on the merged base class data instead of using a meta-learning strategy to learn a robust base feature extractor. Extensive experimentations on three public remote sensing benchmark datasets show that our proposed method can achieve a competitive or state-of-the-art (SOTA) classification performance.**

***Index Terms*— Class surrogate-based supervised contrastive learning (CSSCL), few-shot leaning, remote sensing image scene classification, spatial affinity attention (SAA).**

## NOMENCLATURE

| | |
|---|---|
| $C_{\text{train}}$ | Set of the base classes. |
| $\lvert C_{\text{train}}\rvert$ | Cardinality of $C_{\text{train}}$. |
| $C_{\text{test}}$ | Set of the novel classes and $C_{\text{train}} \bigcap C_{\text{test}} = \emptyset$. |
| $D_{\text{train}}$ | Training dataset and contains sufficient labeled samples from the base classes. |
| Support set | Labeled samples from the novel classes. |
| Query set | Samples from the novel classes staying to be classified. |
| $\tilde{b}_i$ | Embedding vector of a given sample $x_i$. |
| $\tilde{z}_i$ | Contrast vector of a given sample $x_i$. |
| $P$ | Set of class surrogates. |
| $\tilde{p}$ | Class surrogate corresponding to the given anchor sample $x_i$. |
| $p_k$ | Class surrogate of the $k$th class. |

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## I. INTRODUCTION

REMOTE sensing scene classification aims to classify unlabeled remote sensing scene images into different semantic categories based on their scenes [1], [2], [3]. It has widespread practical applications, including disaster detection [4], [5], [6], [7], urban planning [8], [9], land use [10], and so on. In the past few years, remote sensing scene classification methods based on convolutional neural networks (CNNs) have achieved great advances relying on large-scale labeled data samples. However, collecting and annotating a large amount of remote sensing data is time-consuming and labor-extensive in many scenarios, especially, annotating scene images needs to be completed by professional experts with domain knowledge of remote sensing. Hence, annotation-efficient remote sensing scene classification has attracted much research attention.

It is well known that human beings have the ability to recognize new objects only from a few samples. Inspired by this, increasing attention has been paid on few-shot learning (FSL). FSL aims to recognize the novel classes with only a few labeled samples (typically 1–5) [11], [12], [13], [14] using the knowledge learned from the base class datasets which contain sufficient labeled samples. Different from conventional classification tasks, the testing classes of FSL are unseen during the training process and only a few labeled samples from the novel classes can be used while testing.

In recent years, researchers have applied FSL to remote sensing scene classification tasks to overcome the limitation of data scarcity and have continuously improved the few-shot remote sensing scene classification (FSRSSC) performance. Enhancing the representative and discriminative ability of features is the key to improving the FSRSSC performance [8], [15].

However, different from natural images, remote sensing images have unique characteristics. They usually contain many

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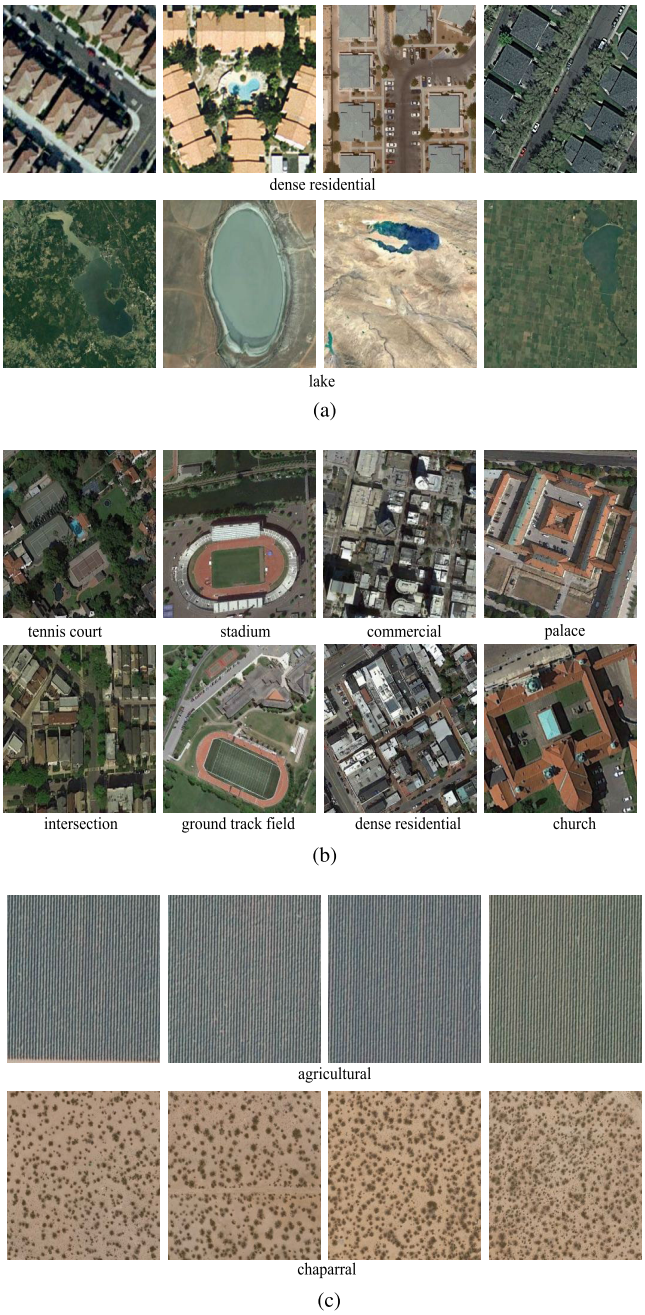



Fig. 1. (a) Large intra-class differences of remote sensing scenes. (b) Large inter-class similarity of remote sensing scenes. (c) High intra-class similarity of remote sensing scenes. These remote sensing images are from the datasets of NWPU-RESISC45 and UC Merced.

scattered and diverse objects owing to the shooting view, making the classification task more challenging. Furthermore, the complex ground objects and uncertain imaging conditions also lead to intra-class differences. Samples of the same semantic category have large visual differences, as shown in Fig. 1(a). Moreover, remote sensing images have inter-class similarity, as shown in Fig. 1(b), "tennis court" and "intersection" are of visual indistinctness and the same can be seen in the other three pairs. In addition, for some specific categories, the remote sensing samples have very high intra-class similarity, as shown in the first row of Fig. 1(c), the scene samples belonging to the category of agricultural have subtle differences. The same is true for the second row of Fig. 1(c). The above characteristics make FSRSSC challenging.

Researchers have explored much to alleviate the above problems and learn representative and discriminative features for remote sensing images. Attention mechanism can enhance representative ability by focusing on important features for the task and suppressing irrelevant ones. So it has been extensively studied to improve the FSRSSC performance. Nevertheless, existing researches mainly use convolution operation to encoding spatial information, which ignores the long-range dependency and the global structural information among spatial locations.

In addition, contrastive learning is also an efficient way to improve the discriminative ability of features, which generally includes self-supervised and fully-supervised manners. Especially, supervised contrastive learning (SCL) [16] leverages label information to contrast samples, achieving more discriminative features. The SCL fulfills representation learning by exploring sample-to-sample relations and a good view is very important for it [17] since too simple or too hard sample pairs both hamper the performance improvement. However, the remote sensing images have the characteristic of large intra-class difference or very high intra-class similarity as mentioned above, which may yield too hard or too simple positive sample pairs as shown in Fig. 1(a) and (c). The complex relations of hard positive pairs may cause positive alignment problem and the too simple positive pairs may wreck the generalization ability of the network model.

To this end, in this article, an end-to-end framework is proposed to improve the performance of FSRSSC via a novel spatial affinity attention (SAA) and a class surrogate-based SCL (CSSCL) strategy.

First, considering that the objects facilitate to understand the scene content, an SAA is proposed to guide the network to focus on the critical regions. It aggregates the mean values and affinities of feature nodes in each channel along the vertical and horizontal directions. Attention maps are generated according to the spatial information. The proposed SAA produces weights for the features in accordance with their channel information and spatial locations, and can model long-range dependency and global structural patterns for the spatial information.

Second, a CSSCL strategy is proposed, which adopts a different way from the SCL to construct positive/negative pairs. Fig. 2 shows the comparison of SCL and CSSCL. SCL construct samples from the same class as positive pairs and those from the different classes as negative pairs. Unlike SCL, CSSCL learns a surrogate for each class in the contrastive space and selects the corresponding surrogate as a positive pair of the anchor. The remaining surrogates and the samples from the different classes are treated as negative pairs. CSSCL solves the positive alignment problem by exploring sample-to-class relations and obtains rich negatives. The feature extractor learned from the base class dataset is applied to novel classes and a logistic regression (LR) classifier is trained using the support set features to classify the query samples.

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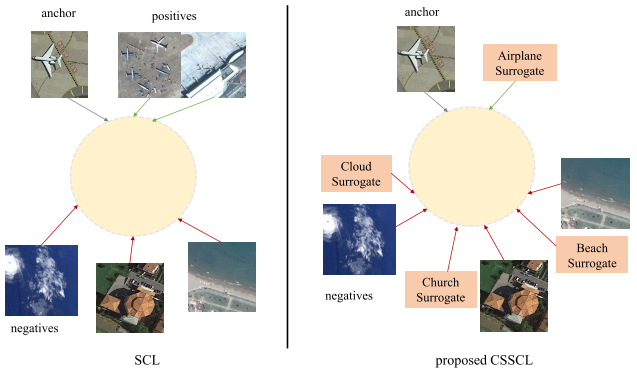



Fig. 2. Comparison of SCL and CSSCL. The main difference of SCL and CSSCL is the way of constructing positive/negative samples. For a specific anchor, SCL contrasts the samples from the same class as positives and the remainder samples of the batch as negatives. Our proposed CSSCL contrasts the corresponding surrogate as positive and the remaining surrogates as well as samples from different classes as negatives.

Our main contributions can be summarized as follows.

1) We propose a novel framework for FSRSSC, which adopts an SAA and a CSSCL strategy to encourage the model to learn representative and discriminative features.
2) We propose an SAA mechanism to adapt to the multiple objects co-occurrence characteristic of the remote sensing images. It guides the network to focus on critical regions by modeling long-range dependency and extracting global structural information, which facilitates to learn more powerful feature representation.
3) A CSSCL strategy is proposed to solve the large intra-class difference and high intraclass similarity problem. It learns a surrogate for each class, and explores both sample-to-class and sample-to-sample relations by pulling the anchors and their corresponding surrogates together and pushing the anchor and the other surrogates as well as the samples from the different classes apart. This helps much for improving the generalization ability of the model.
4) Extensive experiments of FSRSSC on three public remote sensing datasets show that our proposed method can achieve competitive and state-of-the-art (SOTA) performance.

The remaining parts are organized as follows. Section II describes the related work about FSRSSC and contrastive learning. Section III describes our proposed method in detail. Section IV reports the extensive experimental results and analysis on three public datasets. Section V draws the conclusions.

## II. Related Work

In the following, we will review the related work about remote sensing scene classification with FSL and contrastive learning.

### A. Remote Sensing Scene Classification With FSL

Recently, FSRSSC researches have achieved promising performance by adopting the basic ideas that are commonly used in FSL. Existing FSL methods are generally divided into two categories: meta-learning-based methods [18], [19] and pretraining-based methods [15], [20].

The meta-learning-based FSL methods aim to learn a meta-learner using a large amount of labeled base class data and adapt it to the novel classes. During meta-training phase, the samples from the base classes are constructed into episodes to form massive meta-tasks that are similar to FSL conditions. And the parameters of the base learner are optimized by completing those meta-tasks. In particular, the metric-based meta-learning methods [12], [13], [14], [21], [22] have received widespread research attention in FSRSSC. Those methods focus on learning an adaptive embedding space and a metric to effectively learn the relationship among samples [23].

Li et al. [24] proposed a discriminative learning scheme, namely the adaptive match network (DLA-MatchNet), which learns discriminative feature representation with attention mechanism and used an adaptive matcher to measure the similarity between support sample and query sample. Cheng et al. [25] designed prototype self-calibration and inter-calibration strategies in the Siamese-prototype network (SPNet) to learn more accurate prototypes. Li et al. [26] constructed an RS-MetaNet by proposing a novel balance loss to maximize the distance between different classes and adopting a learnable distance metric to measure the similarity. Liu et al. [27] performed ranking-preserving knowledge distillation to enforce the student model to rank support images in the same way as teacher models, achieving better generalization ability. Cui et al. [28] integrated a parametric linear classifier into a meta-learning framework, making the metric more flexible. They also assembled a meta-kernel strategy and a stretching loss to learn discriminative features. Zhang et al. [29] firstly represented each remote sensing image or each scene class as a scene graph and developed a graph matching module to evaluate the similarity between different samples.

The pretraining-based methods obtain a feature extractor on base classes and then transfer it into novel classes. Those methods give over the episodic learning strategy and rely on large-scale data and a good network architecture to extract powerful features with good generalization capability. Many pretraining-based FSRSSC approaches have been proposed in recent years.

Zeng et al. [30] proposed a prototype calibration with a feature-generation model. Wang et al. [31] used shared class sparse principal component analysis (SparsePCA) to settle the negative transfer problem and adopted self-supervised learning (SSL) to improve the robustness of the feature extractor. Xiong et al. [32] utilized weighted Nuclear-norm regularization to boost the FSRSSC performance. They provided random-weighting and self-weighting schemes separately to improve the regularization. Gong et al. [8] proposed a novel two-path aggregation attention mechanism and a novel quad-patch data augmentation for FSRSSC.

Our proposed method belongs to the pretraining-based pipeline.

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### B. Contrastive Learning

Contrastive learning has first been proposed by Hadsell et al. [33]. At present, it has been widely applied to SSL [34], [35] to improve the generalization capability with the idea of pulling positive samples together and pushing negative samples apart.

Khosla et al. [16] extended the self-SCL (SSCL) [35] to a supervised manner. They leveraged label information to construct sample pairs, yielding many positive pairs and accurate negative pairs for an anchor.

Recent research studies [20], [36], [37], [38] introduced contrastive learning into FSL as an additional auxiliary task. Liu et al. [36] investigated noise contrastive estimation (NCE) in meta-learning. Ouali et al. [37] proposed a novel attention-based spatial contrastive learning objective function and applied it to training process to learn the locally discriminative features. Yang et al. [38] integrated contrastive learning into the pre-training and meta-learning stage seamlessly. Specifically, for pre-training, they proposed global and local self-supervised contrastive losses. For meta-learning, they introduced a cross-view episodic training mechanism and a distance-scaled contrastive loss to avoid extreme bias.

Liu et al. [39] proposed a refined prototypical contrastive learning (RPCL) network for few-shot hyperspectral image classification tasks, which incorporates SCL into a prototypical based network. Liu et al. [40] proposed a deep contrastive learning network (DCLN) which uses the pseudo-labeled samples with higher confidence as well as the labeled samples to train a small-sample hyperspectral image classification network. Ning et al. [41] designed an instance-to-instance contrastive learning framework based on category matching for cross-scene hyperspectral image classification tasks.

In the FSRSSC field, contrastive learning has also attracted much research attention. Li et al. [42] weaved SSCL into a prototype-based method to learn powerful features. Gong et al. [43] introduced SSL into meta-learning to solve distribution shifted problem. Ji et al. [44] introduced SCL into FSRSSC. Li et al. [45] trained multi-models using different combinations of self-supervised tasks, which makes full use of the features extracted from different models by ensemble learning. Ji et al. [9] incorporated two auxiliary SCL branches, namely context-guided contrastive learning branch and detail-guided contrastive learning branch, to address the challenges of small inter-class variances and large intra-class variances.

Compared to SSCL, SCL leverages label information to construct positive/negative pairs, generating more accurate image pairs and achieving superior classify accuracy. The experimental results in [44] suggest that SCL can improve performance of FSRSSC. However, remote sensing scene images have the characteristics of low/high intra-class similarity, it may generate accurate but low-quality image pairs when using label information to construct image pairs. This will hinder the model's performance.

Therefore, in this article, a CSSCL strategy is proposed, which selects the corresponding class surrogate instead of the samples from the same class as positive pairs for each anchor and those from the different classes as well as the remaining surrogates as negative pairs. A good embedding space can be learned by pulling positive samples together and pushing negative ones apart. The experimental results demonstrate that CSSCL can achieve a better performance than SCL in FSRSSC tasks.

## III. Proposed Method

In this section, our proposed method will be described in detail, including the formulation of the FSRSSC problem, the overview of our method, feature learning with SAA, CSSCL and few-shot evaluation.

### A. Problem Formulation

Generally, FSRSSC includes training phase and testing phase. During the training phase, a base leaner is trained on the base class dataset $D_{\text{train}} = \{(x_i, y_i)\}$, which consists of the samples from the base classes $C_{\text{train}}$. Here $x_i$ is an remote sensing image and $y_i \in \{0, 1\}^{|C_{\text{train}}|}$ is its one-hot encoded label, $|C_{\text{train}}|$ is the cardinality of $C_{\text{train}}$. At the testing phase, the base learner is tested on the novel classes $C_{\text{test}}$, especially, $C_{\text{train}} \bigcap C_{\text{test}} = \emptyset$. In common, some labeled samples from the novel classes are needed, called support set. The other samples from the novel classes to be classified are defined as query set. Recognizing the query set samples is the goal of FSRSSC. Formally, way and shot are used to formalize the FSRSSC task setting, such as $N$-way, $K$-shot, which means there are $N$ randomly selected novel classes and each class has $K$ support samples during one test process. $K = 1$ or 5, and $N = 5$ are the mainstream settings.

We summarize the main notations and symbols that are used throughout the article in Nomenclature.

### B. Overview

Our proposed framework is shown in Fig. 3. In the training phase, a multi-head network is trained with a shared backbone using the base training set. Following previous works [15], [20], [28], [44], [46], ResNet12 is adopted as the backbone network, $f_\vartheta$, parameterized with $\vartheta$. The network consists of four residual blocks, and each block has three convolutional layers with $3 \times 3$ kernel. Our ResNet12 is identical to that used in [15], [20], [28], [44], and [46], which applies Drop-block [47] as a regularization technique and changes the number of filters from (64, 128, 256, 512) to (64, 160, 320, 640). Our proposed SAA is inserted after each residual block. The backbone, $f_\vartheta$, acts as a feature extractor, which projects an image into an embedding vector with 640 dimensions. Our network architecture also contains two parallel heads $f_\varphi$, $f_\mu$, parameterized with $\varphi$ and $\mu$, respectively, to project the embedding results of $f_\vartheta$ into label space and contrastive space. $f_\varphi$ is a classification function with a single linear layer. And the $f_\mu$ is instantiated with a multilayer perceptron which consists of a single hidden layer with the size of 64. For each output of $f_\mu$, normalization is performed. Especially, the parameters of the backbone, $f_\vartheta$, and the classifier, $f_\varphi$, are jointly optimized by minimizing cross-entropy (CE) loss. The CE loss is defined as follows:

$$L_{\text{CE}} = -\sum_i \log \frac{\exp\left(\tilde{y}^u_{i:y^u_i=1}\right)}{\sum_v \exp\left(\tilde{y}^v_i\right)} \tag{1}$$

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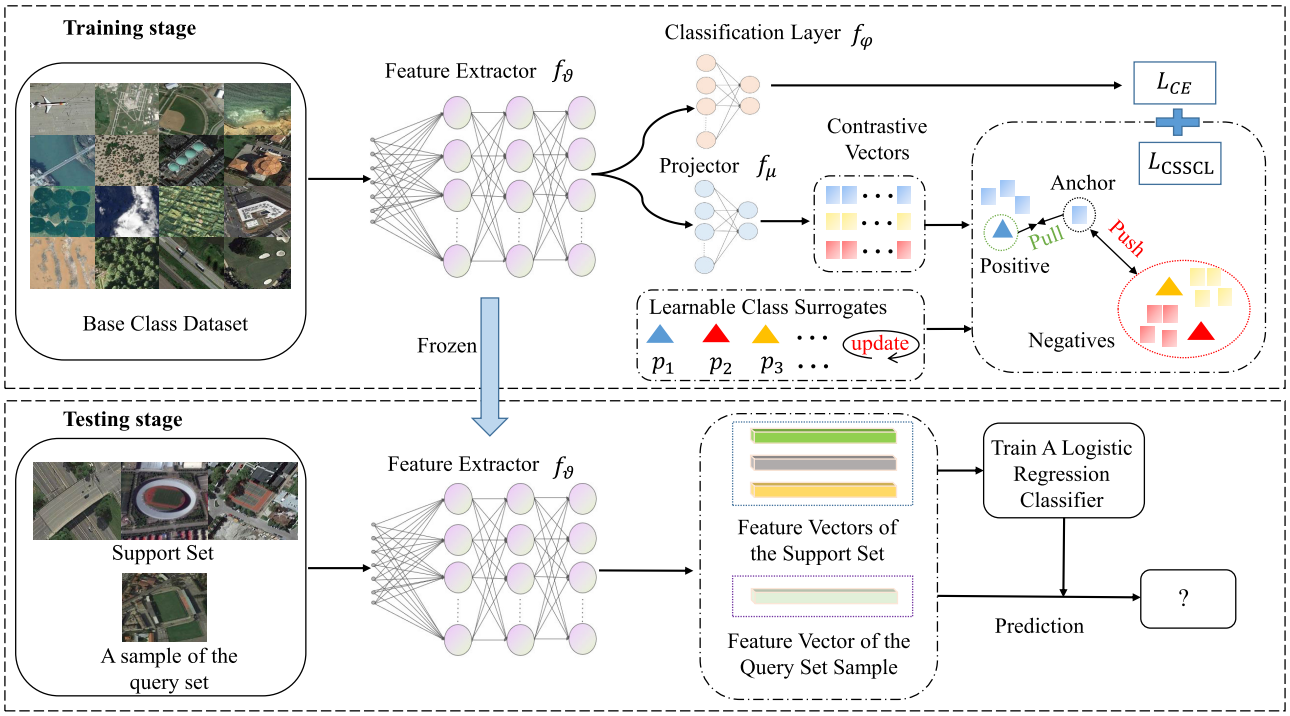



Fig. 3. Framework of our proposed method. An embedding model is trained using the base class dataset. During the testing phase, the trained embedding model is used as a feature extractor and the parameters of the model are frozen.

where $i \in I \equiv \{1, \ldots, \text{Num}\}$ is the index of the samples within the mini-batch, $y_i \in \{0,1\}^{|C_{\text{train}}|}$, $\tilde{y}_i = f_{\vartheta,\varphi}(x_i)$. The optimization function of $f_\mu$ is detailed in Section III-D.

In the testing phase, the parameters of $f_\vartheta$ are frozen and the backbone is used as a feature extractor. The details of testing process is shown in Section III-E.

### C. Feature Learning With SAA

To tackle the co-occurrence of multiple objects problem, an SAA is designed and introduced into the feature learning process to prompt the model to learn more powerful feature representation. The SAA can take any intermediate feature map $F \in \mathbb{R}^{C\times H\times W}$ as input and outputs a refined feature representation $F' \in \mathbb{R}^{C\times H\times W}$ with the same size as $F$. It infers attention maps both channel-wise and spatial-wise, as illustrated in Fig. 4. In the following part, it will be described in detail.

Given the intermediate feature map $F \in \mathbb{R}^{C\times H\times W}$, we first encode the mean values in each channel along the vertical and horizontal directions, respectively, using two spatial extends of pooling kernels $(H, 1)$ or $(1, W)$.

For the specific channel $c$, the output of the height $h$ can be written as

$$a_c^h(h) = \frac{1}{W} \sum_{0\le i<W} f_c(h, i). \tag{2}$$

Similarly, the output of the width $w$ can be written as

$$a_c^w(w) = \frac{1}{H} \sum_{0\le j<H} f_c(j, w). \tag{3}$$

Then, the affinities are computed between all the feature nodes along the two spatial directions to capture the global structural information.

Mathematically, in channel $c$, there are $H$ feature nodes from all the positions along the vertical direction. For the feature node $f_c^h(h) \in \mathbb{R}^W$ at height $h$, compute the affinities of it with all the $H$ feature nodes. Specifically, the pairwise affinity $r_c^h(h, i)$ from the feature node $f_c^h(h)$ to the feature node $f_c^h(i)$ is formulated as

$$r_c^h(h, i) = \theta\left(f_c^h(h)\right)^T \phi\left(f_c^h(i)\right) \tag{4}$$

where $\theta$ and $\phi$ are two transformation functions, formulated as $\theta(f_c^h(h)) = \text{ReLU}(\text{Conv}_\theta^{1\times1}(f_c^h(h)))$, $\phi(f_c^h(h)) = \text{ReLU}(\text{Conv}_\phi^{1\times1}(f_c^h(h)))$, where $\text{Conv}_\theta^{1\times1}$ and $\text{Conv}_\phi^{1\times1}$ are two $1 \times 1$ convolution operations. Similarly, the pairwise affinity $r_c^h(i, h)$ from the feature node $f_c^h(i)$ to the feature node $f_c^h(h)$ is formulated as

$$r_c^h(i, h) = \theta\left(f_c^h(i)\right)^T \phi\left(f_c^h(h)\right). \tag{5}$$

The affinity matrix $\mathbb{R}_c^h \in \mathbb{R}^{H\times H}$ is used to represent the pairwise relations for all the $H$ feature nodes along the vertical direction, where $R_c^h(h, i) = r_c^h(h, i)$. $R_c^h(h, :)$ denotes the $h$th row of $R_c^h$ stacking the pairwise affinities from the feature node $f_c^h(h)$ to all the $H$ vertical directional feature nodes. And $R_c^h(:, h)$ denotes the $h$th column of $R_c^h$ stacking the pairwise affinities from all the $H$ vertical directional feature nodes to the feature node $f_c^h(h)$. The pair $(r_c^h(h, i), r_c^h(i, h))$ represents the bilateral relations between $f_c^h(h)$ and $f_c^h(i)$. $r_c^h(h)$ represents the relation between the feature node $f_c^h(h)$ and all the $H$ vertical directional feature nodes and it is formulated as

$$r_c^h(h) = \frac{1}{2H} \sum_{0\le k<H} \left(R_c^h(h, k) + R_c^h(k, h)\right). \tag{6}$$

Similarly, the pairwise affinity $r_c^w(w, j)$ from the $w$th width feature node $f_c^w(w)$ to the $j$th width feature node $f_c^w(j)$ is

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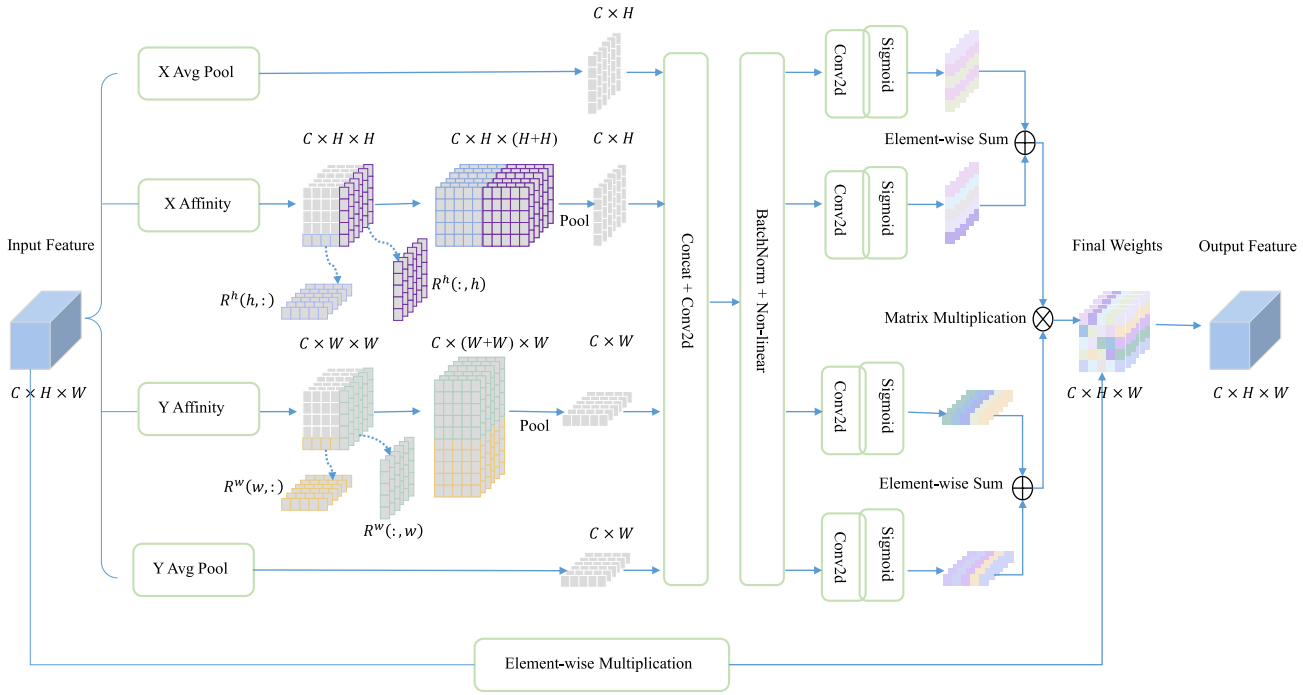



Fig. 4. Overview of SAA. It encodes the mean values of the features and the affinities of the current feature node with all the feature nodes in each channel along the vertical and horizontal directions.

formulated as

$$r_c^w(w, j) = \theta\left(f_c^w(w)\right)^T \phi\left(f_c^w(j)\right). \tag{7}$$

The pairwise affinity $r_c^w(j, w)$ from the feature node $f_c^w(j)$ to the feature node $f_c^w(w)$ is formulated as

$$r_c^w(j, w) = \theta\left(f_c^w(j)\right)^T \phi\left(f_c^w(w)\right). \tag{8}$$

The affinity matrix $R_c^w \in \mathbb{R}^{W\times W}$ is used to represent the pairwise relations for all the $W$ feature nodes along the horizontal direction, where $R_c^w(w, j) = r_c^w(w, j)$. The bilateral relations between $f_c^w(w)$ and $f_c^w(j)$ are represented by the pair $(r_c^w(w, j), r_c^w(j, w))$. The relation between the feature node $f_c^w(w)$ and all the $W$ horizontal directional feature nodes is formulated as

$$r_c^w(w) = \frac{1}{2W} \sum_{0\le l<W} \left(R_c^w(w, l) + R_c^w(l, w)\right). \tag{9}$$

The above transformations yield four direction-aware feature maps $a^h \in \mathbb{R}^{C\times H}$, $a^w \in \mathbb{R}^{C\times W}$, $r^h \in \mathbb{R}^{C\times H}$ and $r^w \in R^{C\times W}$. Next, we will infer attention map using those feature maps.

Specifically, the feature maps generated by (2), (3), (6), and (9), are concatenated together. And then, the convolution operation of filter size $1 \times 1$ followed by a non-linear activation function is conducted on the concatenated features. It can be formulated as follows:

$$\hat{m} = \delta(\text{Conv}^{1\times1}([a^h, a^w, r^h, r^w])) \tag{10}$$

where $\hat{m} \in \mathbb{R}^{C/re\times(2\times H+2\times W)}$ is the intermediate feature map, $re$ is the reduction ratio, $\text{Conv}^{1\times1}$ is a $1 \times 1$ convolution operation, and $\delta$ is a non-linear activation function. Then $\hat{m}$ is split into four separate sub-intermediate maps along the spatial dimension, denoted as $m_a^h \in \mathbb{R}^{C/re\times H}$, $m_a^w \in \mathbb{R}^{C/re\times W}$, $m_r^h \in \mathbb{R}^{C/re\times H}$, $m_r^w \in \mathbb{R}^{C/re\times W}$, respectively. A convolution operation of filter size $1 \times 1$ followed by a sigmoid activation function is conducted on $m_a^h \in \mathbb{R}^{C/re\times H}$ and $m_a^w \in \mathbb{R}^{C/re\times W}$ to yield the attention weights according to the mean values. The operation is formulated as

$$g_a^h = \sigma\left(\text{Conv}_{ah}^{1\times1}\left(m_a^h\right)\right) \tag{11}$$

$$g_a^w = \sigma\left(\text{Conv}_{aw}^{1\times1}\left(m_a^w\right)\right) \tag{12}$$

where $\text{Conv}_{ah}^{1\times1}$ and $\text{Conv}_{aw}^{1\times1}$ are two $1 \times 1$ convolution operations that are used to transform $m_a^h \in \mathbb{R}^{C/re\times H}$ and $m_a^w \in \mathbb{R}^{C/re\times W}$ to tensors with the same channel number as the input intermediate feature map $F$. And $\sigma$ is the sigmoid function. Similarly, a convolution operation of filter size $1 \times 1$ followed by a sigmoid function is conducted on $m_r^h \in \mathbb{R}^{C/re\times H}$ and $m_r^w \in \mathbb{R}^{C/re\times W}$ to yield the attention weights according to the affinities. The operation is written as

$$g_r^h = \sigma\left(\text{Conv}_{rh}^{1\times1}\left(m_r^h\right)\right) \tag{13}$$

$$g_r^w = \sigma\left(\text{Conv}_{rw}^{1\times1}\left(m_r^w\right)\right) \tag{14}$$

where $\text{Conv}_{rh}^{1\times1}$ and $\text{Conv}_{rw}^{1\times1}$ are two $1 \times 1$ convolution operations that are used to transform $m_r^h \in \mathbb{R}^{C/re\times H}$ and $m_r^w \in \mathbb{R}^{C/re\times W}$ to tensors with the same channel number to the input intermediate feature map $F$. Finally the output of the SAA module is represented as

$$f_c'(i, j) = f_c(i, j) \times \left(g_a^h(i) + g_r^h(i)\right) \times \left(g_a^w(j) + g_r^w(j)\right). \tag{15}$$

Our proposed SAA infers attention map along channel and spatial dimensions simultaneously, then the attention map is multiplied to the intermediate feature map to yield an enhanced feature map. It encodes both mean values and affinities in

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each channel along two separate spatial directions, which helps much in modeling long-range dependency and exploiting global structural information. As a result, the SAA can guide the network to focus on the critical regions and extract more powerful features.

### D. Class Surrogate-Based Supervised Contrastive Learning

Compared to SSCL, SCL can construct more accurate positive/negative pairs by leveraging the label information. For each anchor, the samples from the same class constitute positive pairs and samples from the remaining mini-batch constitute negative pairs. Given the sample $x_i$, first map it to an embedding vector $\tilde{b}_i = f_\vartheta(x_i) \in \mathbb{R}^d$, where $d$ is the embedding vector dimension. Then $\tilde{b}_i$ is further mapped to a contrastive vector $\tilde{z}_i = f_\mu(\tilde{b}_i) \in \mathbb{R}^{d'}$, where $d'$ is the contrastive vector dimension. Let $i \in I \equiv \{1, \ldots, \text{Num}\}$ be the index of the samples within the mini-batch. The SCL loss is defined as

$$L_{\text{SCL}} = \sum_{i \in I} \frac{-1}{|B(i)|} \sum_{b \in B(i)} \log \frac{\exp(\tilde{z}_i \tilde{z}_b / \tau)}{\sum_{e \in E(i)} \exp(\tilde{z}_i \tilde{z}_e / \tau)} \tag{16}$$

where $E(i) \equiv I \setminus \{i\}$, $B(i) \equiv \{b \in E(i) : y_b = y_i\}$ is the index of the positives for $x_i$, $|B(i)|$ is the cardinality of $B(i)$. $\tilde{z}_i \tilde{z}_b$ means the inner product of the two contrastive vectors. $\tau$ is a scalar temperature and set to 0.1 in this work.

Designing good positive pairs is one of the keys of contrastive learning. However, as mentioned above, remote sensing scene images may produce too simple or too hard positive pairs due to its low/high intra-class similarity. Such positive pairs are trivial for gaining model's generalization ability. The performance is not satisfied using SCL to solve FSRSSC problem due to poor positive pairs.

To tackle the issue of poor positive pairs, in this article, we propose a CSSCL strategy that can effectively form proper positive pairs by exploring sample-to-class relations. The main idea is to learn a class surrogate $p \in \mathbb{R}^{d'}$ for each class in the contrastive space, where the class surrogate dimension is the same as the contrastive vector dimension. When there are $|C_{\text{train}}|$ base classes, the class surrogates $P \in \mathbb{R}^{|C_{\text{train}}| \times d'}$ would contain $|C_{\text{train}}|$ entries. The class surrogates are used to represent the classes. For the anchor sample $x_i$, CSSCL selects the corresponding class surrogate as positive and the samples from the different classes as well as the remaining surrogates as negatives. The CSSCL loss is defined as

$$L_{\text{CSSCL}} = -\sum_{i \in I} \log \frac{\exp(\tilde{z}_i \tilde{p} / \tau)}{\sum_{k=1}^{|C_{\text{train}}|} \exp(\tilde{z}_i p_k / \tau) + \sum_{e \in E(i)} \exp(\tilde{z}_i \tilde{z}_e / \tau)} \tag{17}$$

where $\tilde{p}$ is the class surrogate corresponding to $x_i$. The parameters of $f_\vartheta$ and $f_\mu$ are jointly optimized by minimizing $L_{\text{CSSCL}}$. The CSSCL pull an anchor and its corresponding surrogate together, and push the anchor away from the negative samples. For each class surrogate $p \in \mathbb{R}^{d'}$, we initialize it in the same way as the parameters of the network and it needs to be learned during the training process. Specifically, the parameters of each class surrogate $p$ can be updated with the following equation:

$$p^{t+1} = p^t - \gamma \frac{\partial L_{\text{CSSCL}}}{\partial p^t} \tag{18}$$

where $\gamma$ is the learning rate.

The final objective function during training phase is obtained by combining the CE loss and the CSSCL loss. The objective function is formulated as follows:

$$L_{\text{train}} = L_{\text{CE}} + \lambda L_{\text{CSSCL}} \tag{19}$$

where $\lambda$ is used to balance $L_{\text{CE}}$ and $L_{\text{CSSCL}}$.

Compared to SCL, our proposed CSSCL adopts a different way to construct positive/negative pairs, which can mitigate the hard or simple positives problem and enrich the negatives at the mean time. It encourages the model to learn more discriminative representation, which facilitates to boost the performance of FSRSSC.

### E. Few-Shot Evaluation

As shown in Fig. 3, the embedding features of both support set samples and query set samples are extracted using the learned model and are normalized by default. A LR classifier is trained using the embedding features of the support set samples and the corresponding labels to infer the labels of the embedding features of the query set samples. The LR classifier is trained by optimizing the following equation [15], [20]:

$$\eta = \arg\min_{\{W,o\}} \sum_{s=1}^{S} L_{\text{CE}}^s(W f_\vartheta(x_s) + o, y_s) + R(W, o) \tag{20}$$

where $W$ is the weight and $o$ is the bias of the LR classifier. $S$ is the number of samples in support set. $R$ is the regularization term.

## IV. Experimental Results and Analysis

To demonstrate the effectiveness of our proposed method, we conduct experiments on three benchmark datasets, namely NWPU-RESISC45 [48], WHU-RS19 [49], and UC Merced Land Use [50]. The proposed method is compared with the SOTA methods and the ablation experiments are conducted. Afterward, we explain how CSSCL increases the feature discriminative ability. We also analyze the impact of parameters and embedding network architecture on classification accuracy. Finally, the influence of different attention mechanisms is explored.

### A. Datasets and Evaluation Criterion

NWPU-RESISC45, WHU-RS19, and UC Merced Land Use are three well-established remote sensing scene classification datasets. For a fair comparison, we split the datasets for training, verification, and testing according to the literatures [8], [24], [27], [28].

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*1) NWPU-RESISC45:* Northwestern Polytechnic University released the NWPU-RESISC45 dataset in 2017, which covers more than 100 countries and regions from Google Earth. This dataset contains 45 scene categories, and each category has 700 images with a size of 256 × 256 pixels. The resolution of most of the images is from 0.2 to 30 m. The classes are split into 25, 10, and 10 for training, verification, and testing.

*2) WHU-RS19:* The WHU-RS19 dataset was released by Wu Han University in 2012. The dataset contains 1005 scene images from 19 categories. There are at least 50 images with the size of 600 × 600 pixels in each category. The dataset is divided into 9, 5, and 5 for training, verification, and testing.

*3) UC Merced:* The UC Merced dataset contains 21 scene categories including urban and rural land types. All the images are from the United States Geological Survey (USGS) National Map. In each category, there are 100 images with a size of 256 × 256 pixels and a spatial resolution of 0.3 m. All the 21 categories are divided into 10, 5, and 6 for training, verification, and testing.

*4) Classification Accuracy:* With reference to the previous works, for fair comparison, classification accuracy (Acc) is used in this article to evaluate the classification performance. Acc is formulated as

$$\text{Acc} = \frac{1}{T}\sum_{ta=1}^{T}\frac{\text{cor}^{ta}}{N \times qu} \tag{21}$$

where $T$ is the total number of FSRSSC tasks, $\text{cor}^{ta}$ is the number of correctly classified query samples in task $ta$, and $qu$ is the number of query samples in each novel class.

## B. Implementation Details

The proposed model is implemented and tested on the Pytorch platform with a NVIDIA RTX 3090 GPU. The stochastic gradient descent (SGD) optimizer is used with a momentum of 0.9 to update the parameters of the models and the class surrogates. The resolution of the images are uniformed to 84 × 84. For data augmentation, we use the augmentations in SimCLR [35] and apply the image rotation operations by 0°, 90°, 180°, and 270° on both original and augmented images. Thus, seven copies of each original image are generated in the mini-batch at each iteration. For experiments on the NWPU-RESISC45 dataset, the learning rate of the network as well as $\gamma$ in (18) are initialized as 0.05 and the weight decay is 5e-4, the total training epochs are set to 90 and the learning rates are decayed one time after 80 epochs. For experiments on WHU-RS19 and UC Merced datasets, the learning rate of the network as well as $\gamma$ in (18) are initialized as 0.01. We train 110 epochs for the experiments on the WHU-RS19 dataset; the learning rates are decayed two times after the first 80 epochs. For experiments on the UC Merced dataset, we train 90 epochs and the learning rates are decayed one time after 80 epochs. The reduction ratio of the SAA is fixed to 32. The balance factor $\lambda$ in (19) are set as 0.1.

For evaluation, the five-way one-shot and five-way five-shot experiments are conducted and our method is evaluated with three runs, where in each run the accuracy is the mean accuracy of 600 randomly sampled episodes. The final mean accuracy and the 95% confidence intervals are reported. The number of query samples in each class is set to 15.

TABLE I
FIVE-WAY CLASSIFICATION ACCURACY ON THE NWPU-RESISC45 DATASET

| Methods | Year | Backbone Network | 5-way 1-shot | 5-way 5-shot |
|---|---|---|---|---|
| MAML [11] | 2017 | ConvNet | $58.99 \pm 0.45$ | $72.67 \pm 0.38$ |
| Matching Net [12] | 2016 | ConvNet | $60.21 \pm 0.77$ | $71.66 \pm 0.45$ |
| ProtoNet [13] | 2017 | ConvNet | $52.77 \pm 0.44$ | $75.32 \pm 0.35$ |
| Relation Net [14] | 2018 | ConvNet | $60.77 \pm 0.47$ | $76.08 \pm 0.34$ |
| MetaOptNet [47] | 2019 | ResNet-12 | $62.72 \pm 0.64$ | $80.41 \pm 0.41$ |
| DLA-MatchNet [24] | 2021 | ConvNet | $68.80 \pm 0.70$ | $81.63 \pm 0.46$ |
| RS-MetaNet [26] | 2021 | ResNet-50 | $52.78 \pm 0.09$ | $71.49 \pm 0.81$ |
| TAE-Net [46] | 2021 | ResNet-12 | $69.13 \pm 0.83$ | $82.37 \pm 0.52$ |
| Meta-FSEO [52] | 2021 | ConvNet | $63.08 \pm 0.48$ | $74.29 \pm 0.43$ |
| MetaLearning [53] | 2021 | ResNet-12 | $69.46 \pm 0.22$ | $84.66 \pm 0.12$ |
| SCL-MLNet [42] | 2022 | ConvNet | $62.21 \pm 1.12$ | $80.86 \pm 0.76$ |
| MKN [28] | 2022 | ResNet-12 | $65.84 \pm 0.89$ | $82.67 \pm 0.55$ |
| MGFF [54] | 2022 | ResNet-12 | $75.09 \pm 0.94$ | $83.24 \pm 0.65$ |
| IDLN [51] | 2022 | ResNet-12 | $75.25 \pm 0.75$ | $84.67 \pm 0.23$ |
| SPNet [25] | 2022 | ResNet-18 | $67.84 \pm 0.87$ | $83.94 \pm 0.50$ |
| DANet [8] | 2022 | ResNet-12 | $74.30 \pm 0.20$ | $87.29 \pm 0.11$ |
| Ji *et al.* [44] | 2022 | ResNet-12 | $76.70 \pm 0.44$ | $89.87 \pm 0.21$ |
| SGMNet [29] | 2022 | ConvNet | $70.40 \pm 0.83$ | $82.32 \pm 0.53$ |
| Multiform Ensemble [45] | 2023 | ResNet-12 | **$79.37 \pm 0.20$** | **$89.94 \pm 0.11$** |
| DCN [9] | 2023 | ResNet-12 | $74.40 \pm 0.78$ | $89.22 \pm 0.41$ |
| Ours | 2023 | ResNet-12 | **$77.96 \pm 0.87$** | **$91.17 \pm 0.51$** |

## C. Comparison Results With Other Methods

To demonstrate the effectiveness of our proposed method, we compare it with 20 representative methods, including MAML [11], MatchingNet [12], ProtoNet [13], DLA-MatchNet [24], RS-MetaNet [26], IDLN [51], Ji et al. [44], SGMNet [29], Multiform Ensemble [45], DCN [9], and so on, where MAML [11], MatchingNet [12], ProtoNet [13], etc., are classical FSL methods and DLA-MatchNet [24], RS-MetaNet [26], IDLN [51], Ji et al. [44], SGMNet [29], Multiform Ensemble [45], DCN [9], etc., are the SOTA FSRSSC methods. The results of the classical FSL methods are cited from Ji et al. [44] and DANet [8] and the results of the SOTA FSRSSC methods from the original published articles, respectively.

The experimental results on three datasets are shown in Tables I–III, respectively, and the optimal two results are highlighted with different colors. It can be seen that our proposed method achieves SOTA performance except in the five-way one-shot task on the NWPU-RESISC45 dataset. Next, we will analyze the results in detail.

*1) NWPU-RESISC45:* Table I shows the comparison results on the NWPU-RESISC45 dataset. Our method achieves the optimal results in the five-way five-shot task and the suboptimal results in the five-way one-shot task. Specifically, in the five-shot setting, our method outperforms the SOTA (multiform ensemble) up to 1.23%, Ji et al. [44] up to 1.3% and SGMNet up to 8.85%. In the one-shot setting, Multiform Ensemble achieves the optimal performance. The reason is that this method adopts a multiform ensemble training strategy. An ensemble model of multiple feature extractors is trained using different SSL methods, and the feature vectors from

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TABLE II
FIVE-WAY CLASSIFICATION ACCURACY ON THE WHU-RS19 DATASET

| Methods | Year | Backbone Network | 5-way 1-shot | 5-way 5-shot |
|---|---|---|---|---|
| MAML [11] | 2017 | ConvNet | $59.92 \pm 0.35$ | $82.30 \pm 0.23$ |
| Matching Net [12] | 2016 | ConvNet | $73.52 \pm 0.35$ | $86.04 \pm 0.20$ |
| ProtoNet [13] | 2017 | ConvNet | $59.52 \pm 0.37$ | $85.58 \pm 0.33$ |
| Relation Net [14] | 2018 | ConvNet | $66.97 \pm 0.35$ | $79.62 \pm 0.21$ |
| MetaOptNet [47] | 2019 | ResNet-12 | $62.05 \pm 0.76$ | $82.60 \pm 0.46$ |
| DLA-MatchNet [24] | 2021 | ConvNet | $68.27 \pm 1.83$ | $79.89 \pm 0.33$ |
| TAE-Net [46] | 2021 | ResNet-12 | $73.67 \pm 0.74$ | $88.95 \pm 0.52$ |
| MGFF [54] | 2022 | ResNet-12 | $76.80 \pm 0.96$ | $84.86 \pm 0.76$ |
| IDLN [51] | 2022 | ResNet-12 | $73.89 \pm 0.88$ | $83.12 \pm 0.56$ |
| SPNet [25] | 2022 | ResNet-18 | $81.06 \pm 0.60$ | $88.04 \pm 0.28$ |
| DANet [8] | 2022 | ResNet-12 | $75.02 \pm 0.16$ | $89.21 \pm 0.07$ |
| Ji *et al.* [44] | 2022 | ResNet-12 | $86.29 \pm 0.30$ | $92.96 \pm 0.12$ |
| SGMNet [29] | 2022 | ConvNet | $85.06 \pm 0.55$ | $90.12 \pm 0.27$ |
| Multiform Ensemble [45] | 2023 | ResNet-12 | **$87.21 \pm 0.12$** | **$93.58 \pm 0.07$** |
| DCN [9] | 2023 | ResNet-12 | $81.74 \pm 0.55$ | $91.67 \pm 0.25$ |
| Ours | 2023 | ResNet-12 | **$87.42 \pm 0.16$** | **$94.67 \pm 0.49$** |

TABLE III
FIVE-WAY CLASSIFICATION ACCURACY ON THE UC MERCED DATASET

| Methods | Year | Backbone Network | 5-way 1-shot | 5-way 5-shot |
|---|---|---|---|---|
| MAML [11] | 2017 | ConvNet | $48.86 \pm 0.74$ | $60.78 \pm 0.62$ |
| Matching Net [12] | 2016 | ConvNet | $46.46 \pm 0.75$ | $67.39 \pm 0.50$ |
| ProtoNet [13] | 2017 | ConvNet | $54.03 \pm 0.78$ | $70.81 \pm 0.48$ |
| Relation Net [14] | 2018 | ConvNet | $49.77 \pm 0.78$ | $57.98 \pm 0.57$ |
| DLA-MatchNet [24] | 2021 | ConvNet | $53.76 \pm 0.62$ | $63.01 \pm 0.51$ |
| RS-MetaNet [26] | 2021 | ResNet-50 | $57.23 \pm 0.56$ | $76.08 \pm 0.28$ |
| TAE-Net [46] | 2021 | ResNet-12 | $60.21 \pm 0.72$ | $77.44 \pm 0.51$ |
| SCL-MLNet [42] | 2022 | ConvNet | $51.37 \pm 0.79$ | $68.09 \pm 0.92$ |
| MKN [28] | 2022 | ResNet-12 | $58.45 \pm 0.54$ | $77.92 \pm 0.32$ |
| SPNet [25] | 2022 | ResNet-18 | $57.64 \pm 0.73$ | $73.52 \pm 0.51$ |
| DANet [8] | 2022 | ResNet-12 | $61.60 \pm 0.18$ | $78.02 \pm 0.16$ |
| SGMNet [29] | 2022 | ConvNet | $60.52 \pm 0.74$ | $73.42 \pm 0.49$ |
| Multiform Ensemble [45] | 2023 | ResNet-12 | **$64.21 \pm 0.17$** | **$80.95 \pm 0.15$** |
| DCN [9] | 2023 | ResNet-12 | $58.64 \pm 0.71$ | $76.61 \pm 0.49$ |
| Ours | 2023 | ResNet-12 | **$65.79 \pm 0.72$** | **$84.16 \pm 0.33$** |

multiple trained feature extractors are concatenated for testing. But Ji et al. [44], SGMNet and our method just trained a single model. In addition, Multiform Ensemble, Ji et al. [44] and our method outperform SGMNet in a large margin, which owes to the deeper backbone network.

*2) WHU-RS19 and UC Merced:* As seen in Tables II and III, our method achieves the optimal results on WHU-RS19 and UC Merced datasets both in five-way five-shot and five-way one-shot tasks. Specifically, on the WHU-RS19 dataset, our method outperforms Multiform Ensemble by 1.09% and 0.21% in the five-way five-shot and five-way one-shot tasks, respectively. On the UC Merced dataset, our method outperforms Multiform Ensemble by 3.21% in five-shot setting and by 1.58% in one-shot setting. The experimental results show the superiority of our method.

### D. Ablation Study

To study the effectiveness of different components in our method, we conduct ablation experiments on the three datasets. RFS-simple [15] is used as the baseline which is trained by the CE loss and adopts Resnet12 as the backbone network. Table IV shows the experimental results. It can be seen that, on the three datasets, the baseline achieves 72.86%, 82.62% and 57.07% in one-shot setting, and 88.76%, 92.45% and 75.91% in five-shot setting.

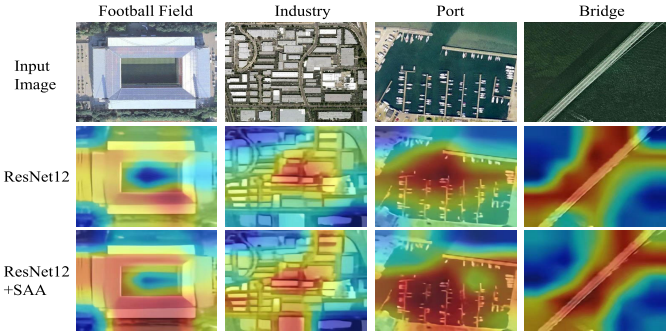



Fig. 5. Heat map visualization. Each column contains the input image, the attention map of baseline model (ResNet12) and that of the model with SAA module (ResNet12 + SAA).

When the SAA is introduced to the backbone network, for the five-way one-shot task, the classification accuracy is improved by 2.99%, 3.58% and 3.30%, respectively, on each dataset. As for the five-way five-shot task, the classification accuracy is improved by 1.74%, 0.94% and 2.57%, respectively. The experimental results demonstrate that the proposed SAA can guide the network to focus on important targets. And, the embedding feature heat map utilizing grad-class activation mapping (CAM) [55] is presented in Fig. 5 to furtherly demonstrate the effectiveness of SAA, where the regions contributing more to the final classification are indicated in darker red color. The column of the Football Field shows that, both the baseline and the model with SAA module can focus on most of the clue areas, but the model with SAA module has a stronger focus on key cues than the baseline model. The column of the Industry shows that the model with SAA module can fully perceive relevant features, while the baseline model loses some clues. The column of the Port shows that the baseline model focuses mainly on the water, while the model with SAA module focuses both on the water and the boats in the port. The column of the Bridge shows that the model with SAA module highlights the bridge while the baseline model highlights the background region. Obviously, the model equipped with SAA can comprehensively and accurately capture key cues and enhance the representation capability of the features.

The addition of the CSSCL loss can improve the classification accuracy by 2.52%, 2.39% and 4.70% on the NWPU-RESISC45, WHU-RS19 and UC Merced datasets, respectively, in one-shot setting. Similarly, the classification accuracy is improved by 0.94%, 1.16% and 5.79% in five-shot setting. The results demonstrate that, with the joint supervision of CE loss and CSSCL loss, the model can extract more discriminative features which facilitates to recognize new classes.

As can be seen in Table IV, significant improvements can be obtained both in one-shot (5.10%, 4.80%, 8.72%) and five-shot (2.41%, 2.22%, 8.25%) tasks by applying SAA and CSSCL strategy simultaneously. The results of ablation study demonstrate that our proposed SAA and CSSCL are effective in FSRSSC tasks.

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TABLE IV
FIVE-WAY CLASSIFICATION ACCURACY ON THE NWPU-RESISC45, WHU-RS19, AND UC MERCED DATASETS OF THE ABLATION STUDY

| *Baseline* | *SAA* | *CSSCL* | NWPU-RESISC45,5-way | | WHU-RS19,5-way | | UC Merced,5-way | |
|---|---|---|---|---|---|---|---|---|
| | | | 1-shot | 5-shot | 1-shot | 5-shot | 1-shot | 5-shot |
| ✓ | | | 72.86 | 88.76 | 82.62 | 92.45 | 57.07 | 75.91 |
| ✓ | ✓ | | 75.85 | 90.50 | 86.20 | 93.39 | 60.37 | 78.48 |
| ✓ | | ✓ | 75.38 | 89.70 | 85.01 | 93.61 | 61.77 | 81.70 |
| ✓ | ✓ | ✓ | **77.96** | **91.17** | **87.42** | **94.67** | **65.79** | **84.16** |

TABLE V
CLASSIFICATION ACCURACY WITH DIFFERENT NOVEL CLASSES ON THE NWPU-RESISC45 DATASET

| Model | 8-way | | 10-way | |
|---|---|---|---|---|
| | 1-shot | 5-shot | 1-shot | 5-shot |
| Baseline | 63.76 | 83.68 | 60.69 | 81.03 |
| Ours | 69.16 | 86.10 | 65.34 | 84.14 |

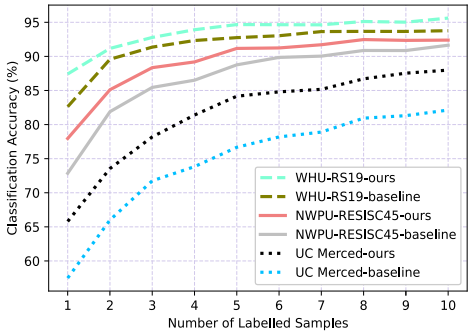



Fig. 6. Classification accuracy on the NWPU-RESISC45, WHU-RS19, and UC Merced datasets with different numbers of labeled samples.

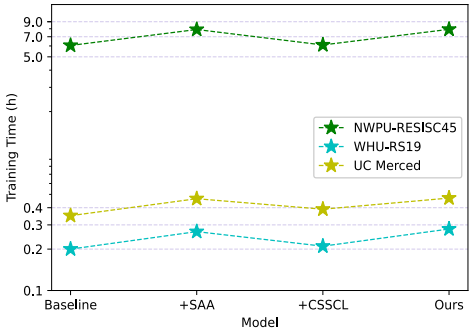



Fig. 7. Training runtime on the NWPU-RESISC45, WHU-RS19, and UC Merced datasets under different methods.

To verify the effectiveness of our proposed method on different numbers of new categories, we also conduct eight-way and ten-way experiments in one-shot and five-shot settings on the NWPU-RESISC45 dataset. Table V shows the experimental results. It can be seen that our proposed method always achieves a better classification performance than that of baseline. Moreover, the classification accuracy of the baseline method and our proposed method with different number of labeled samples is shown in Fig. 6. We can see that the classification accuracy of the two methods increases as the number of labeled samples increases on the three datasets. Our proposed method outperforms the baseline method consistently with different numbers of labeled samples.

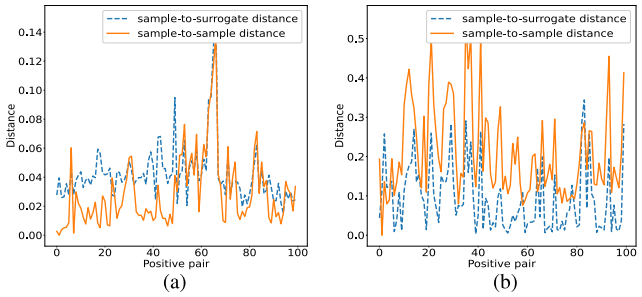



Fig. 8. Sample-to-surrogate and sample-to-sample distances of chaparral and dense residential. (a) Chaparral. (b) Dense residential.

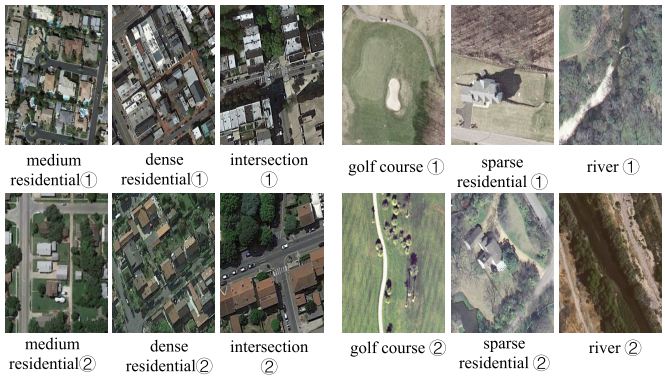



Fig. 9. Hard samples which have large intra-class visual differences and high inter-class similarity. Samples of the left and right group come from the NWPU-RESISC45 and UC Merced dataset, respectively.

Fig. 7 presents the training runtime of different methods on the three datasets. We can see that the maximum training time of our proposed approach is less than 9 h, even though it takes a slightly more time than that of the baseline method. In the inference stage, we use a LR classifier trained using the support set to inference the queries. It takes 0.13 and 0.4 s, respectively, on our proposed method and 0.11 and 0.33 s, respectively, on the baseline method to inference the 15 queries of per task in one-shot and five-shot settings on the three datasets. Moreover, the FLOPs of our proposed model is 3.68G and that of ResNet-12 is 3.52G.

### *E. How CSSCL Increases the Feature Discriminative Ability*

To further understand the CSSCL, we visualize the distances of positive pairs. Fig. 8 shows the sample-to-surrogate and sample-to-sample distances of two classes (Chaparral and Dense residential) from the UC Merced dataset at intermediate training stage. The sample-to-surrogate distance means the cosine distance between the anchor sample and its corresponding class surrogate. And the sample-to-sample distance means the cosine distance between the anchor

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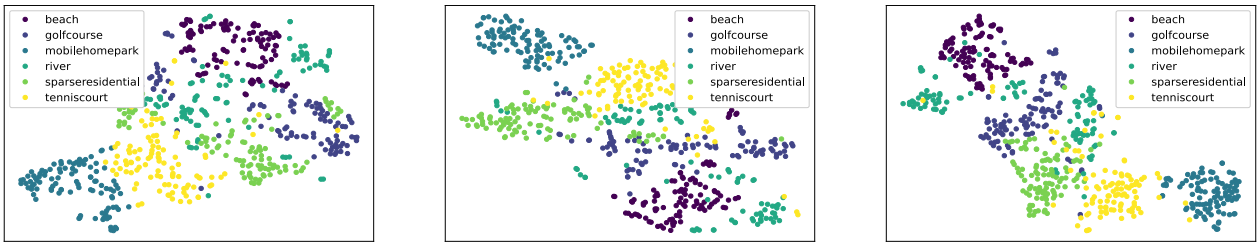



Fig. 10. Visualization of the novel classes embedding from UC Merced dataset. The features are from six classes with 100 samples per class and presented with different colors. (a) Embedding from the baseline model. (b) Embedding from the SCL model. (c) Embedding from our proposed model.

sample and its positive sample selected from the same class. In Fig. 8(a), it can be found that the sample-to-surrogate distances are generally larger than the sample-to-sample distances. And in practical, the images of chaparral are of similar appearances as shown in Fig. 1(b), indicating that samples of chaparral have the high intra-class similarity. The larger sample-to-surrogate distances suggest that our proposed CSSCL can ease the too easy positive pair problem caused by extremely high intra-class similarity. In Fig. 8(b), it can be found that the sample-to-sample distances are much larger than the sample-to-surrogate distances. And it can be also observed that in Fig. 1(a), the images of dense residential are of dissimilar appearance, indicating that samples of dense residential have the lower intra-class similarity. The smaller sample-to-surrogate distances suggest that our proposed CSSCL can mitigate the too hard positive pair problem caused by low intra-class similarity.

To confirm the superiority of the CSSCL, we compare it with SCL. The comparison results on the three datasets in the two FSL settings are shown in Table VI. +SCL and +CSSCL mean introduce the SCL loss and the CSSCL loss into the baseline, respectively. It can be seen from Table VI that, CSSCL outperforms SCL in all the experiments of the three datasets. Especially, CSSCL achieves improvements with a large margin on the UC Merced dataset both in one-shot setting (4.70%) and five-shot setting (5.79%). The comparison results demonstrate that our proposed CSSCL strategy can mitigate the hard or simple positives problem to a certain extent that occurs when the SCL strategy is utilized. Moreover, we select hard samples from the NWPU-RESISC45 and UC Merced test dataset and compare the classification performances of CSSCL and SCL on those hard samples. Fig. 9 shows some examples of the hard samples which are difficult to distinct due to the intra-class visual differences and inter-class similarity. The hard samples come from three categories within each dataset and the classify accuracies in three-way one-shot and three-way five-shot settings are shown in Table VII. The comparison results in Table VII furtherly indicate the superiority of the CSSCL.

To visually show the discriminative ability of the features, we visualize the embedding vectors extracted by three models using t-SNE [56]: the baseline model, the SCL model that trained with joint supervision of CE loss and SCL loss, and our proposed model. Fig. 10 shows the feature distribution of novel class samples in the UC Merced dataset. From Fig. 10(a), it can be observed that the features extracted by the baseline model are not separable very well. The intra-class distances

TABLE VI
CLASSIFICATION ACCURACY OF SCL AND CSSCL ON THE NWPU-RESISC45, WHU-RS19, AND UC MERCED DATASETS

| Model | NWPU-RESISC45 5-way | | WHU-RS19 5-way | | UC Merced 5-way | |
|---|---|---|---|---|---|---|
| | 1-shot | 5-shot | 1-shot | 5-shot | 1-shot | 5-shot |
| Baseline | 72.86 | 88.76 | 82.62 | 92.45 | 57.07 | 75.91 |
| +SCL | 73.71 | 89.06 | 84.42 | 93.03 | 58.81 | 77.83 |
| +CSSCL | 75.38 | 89.70 | 85.01 | 93.61 | 61.77 | 81.70 |

TABLE VII
CLASSIFICATION ACCURACY OF SCL AND CSSCL ON HARD SAMPLES

| Model | NWPU-RESISC45,3-way | | UC Merced,3-way | |
|---|---|---|---|---|
| | 1-shot | 5-shot | 1-shot | 5-shot |
| Baseline | 72.77 | 86.06 | 55.32 | 75.11 |
| +SCL | 73.86 | 86.52 | 56.04 | 77.48 |
| +CSSCL | 74.68 | 87.70 | 61.23 | 79.84 |

of the features are large and the boundaries between classes are not distinct. Fig. 10(b) shows the features extracted by the SCL model, the intra-class features are more compact and the inter-class features are more separable compared to the baseline model. Fig. 10(c) shows that our proposed method can furtherly improve the intra-class compactness and inter-class dispersion of the deep features, which facilitates to improve the classification accuracy significantly and demonstrates the effectiveness of our proposed CSSCL.

### F. Parameter Analysis

*1) Impact of Parameter $\lambda$ on Classification Accuracy:* In this section, we investigate the influence of the weight coefficient $\lambda$ in (19) on the overall classification performance, which balances the CE loss and the CSSCL loss. In the experimental process, we vary $\lambda$ in the range of $[0, 0.01, 0.05, 0.1, 0.5, 1]$. The accuracies in one-shot and five-shot settings on the three datasets are shown in Fig. 11. It can be observed that the optimal performance can be achieved under both settings when $\lambda = 0.1$. Therefore, in this article, we choose 0.1 as the weight coefficient of CSSCL loss.

*2) Impact of Parameter Reduction Ratio on Classification Accuracy:* Table VIII shows the effect of different reduction ratio on the classification performance. The optimal reduction ratio is different for different datasets. For the NWPU-RESISC45 dataset, the performance achieves optimal when the reduction ratio is 32. For the WHU-RS19 dataset, the

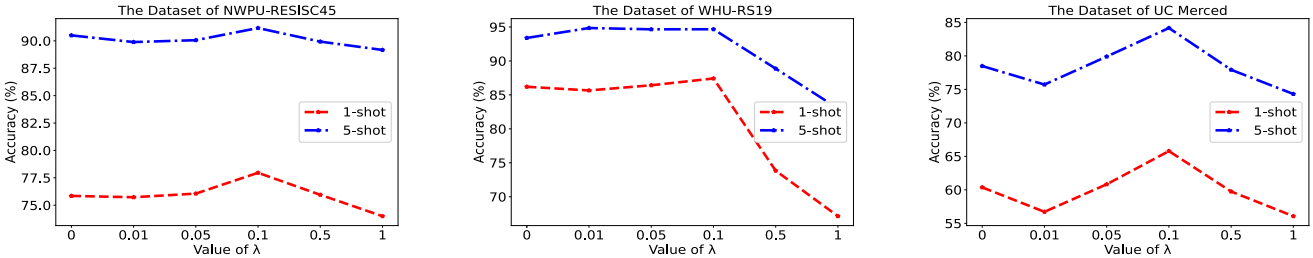
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TABLE VIII
CLASSIFICATION ACCURACY WITH DIFFERENT REDUCTION RATIO ON THE NWPU-RESISC45, WHU-RS19, AND UC MERCED DATASETS

| Reduction Ratio | NWPU-RESISC45,5-way | | WHU-RS19,5-way | | UC Merced,5-way | |
|---|---|---|---|---|---|---|
| | 1-shot | 5-shot | 1-shot | 5-shot | 1-shot | 5-shot |
| 4 | 75.15 | 90.45 | 83.98 | 93.33 | 58.45 | 77.07 |
| 8 | 74.37 | 90.31 | 85.33 | 93.85 | 60.50 | 80.09 |
| 16 | 74.65 | 89.97 | 85.27 | 93.86 | 60.04 | 77.99 |
| 32 | 75.85 | 90.50 | 86.20 | 93.39 | 60.37 | 78.48 |

TABLE IX
CLASSIFICATION ACCURACY OF DIFFERENT BACKBONES ON THE NWPU-RESISC45, WHU-RS19, AND UC MERCED DATASETS

| Backbone | NWPU-RESISC45,5-way | | WHU-RS19,5-way | | UC Merced,5-way | |
|---|---|---|---|---|---|---|
| | 1-shot | 5-shot | 1-shot | 5-shot | 1-shot | 5-shot |
| ConvNet4-64 | 69.21 | 85.33 | 81.41 | 91.51 | 57.05 | 75.05 |
| ConvNet4-256 | 72.21 | 87.29 | 82.47 | 92.73 | 61.51 | 79.95 |
| ResNet-12 | 77.96 | 91.17 | 87.42 | 94.67 | 65.79 | 84.16 |
| ResNet-18 | 74.71 | 89.17 | 83.31 | 92.33 | 59.93 | 78.19 |




(a) (b) (c)

Fig. 11. Impact of weight coefficient $\lambda$ on three datasets. Results on (a) NWPU-RESISC45 dataset, (b) WHU-RS19 dataset, and (c) UC Merced dataset.

TABLE X
CLASSIFICATION ACCURACY OF DIFFERENT ATTENTION MODULES ON THE NWPU-RESISC45, WHU-RS19, AND UC MERCED DATASETS

| Model | NWPU-RESISC45,5-way | | WHU-RS19,5-way | | UC Merced,5-way | |
|---|---|---|---|---|---|---|
| | 1-shot | 5-shot | 1-shot | 5-shot | 1-shot | 5-shot |
| Baseline | 72.86 | 88.76 | 82.62 | 92.45 | 57.07 | 76.89 |
| +Channel Attention(SE) | 74.67 | 89.32 | 84.95 | 93.46 | 58.44 | 76.25 |
| +CBAM | 74.94 | 89.49 | 84.69 | 93.17 | 58.41 | 76.51 |
| +Coordinate Attention | 75.33 | 89.88 | 85.35 | **93.61** | 59.43 | 77.49 |
| +SAA (Ours) | **75.85** | **90.50** | **86.20** | 93.39 | **60.37** | **78.48** |

optimal values for one-shot setting and five-shot setting are 32 and 16, respectively. For the UC Merced dataset, the optimal value is 16. Taking computational complexity and classification accuracy into account, the reduction ratio is set as 32 in this article.

### *G. Effect of Embedding Network Architecture on Classification Accuracy*

To study the influence of embedding network architecture on the classification accuracy, we conduct comparison experiments using four different backbones: ConvNet4-64 [13], ConvNet4-256, ResNet-18 [25], and ResNet-12. ConvNet4-64 and ConvNet4-256 are both composed of four convolutional blocks. Each block contains a $3 \times 3$ convolution, a batch normalization layer, a ReLU nonlinearity and a $2 \times 2$ max-pooling layer. The ConvNet4-64 finally produces a 64-D feature vector and the dimension of the feature vector produced by ConvNet4-256 is 256. Table IX shows the comparison results of different backbones on the three datasets. We can find that ResNet-12 achieves the best performance. The performance degrades if the backbone is too complex due to the overfitting problem. So in this article, ResNet-12 is selected as the backbone network.

### *H. Comparison With Other Attention Modules*

In this section, we compared the SAA with other commonly used attention mechanisms, including channel attention (CA [57]), CBAM [58] and Coordinate attention [59]. Table X shows the comparison results on the three datasets in two FSL situations. +channel attention (CA), +CBAM, +Coordinate attention, +SAA mean CA, CBAM, Coordinate attention,

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SAA are embedded after each of the four residual blocks in ResNet12, respectively.

As shown in Table X, our proposed SAA achieves the optimal classification performance improvement except in the five-way five-shot task of the WHU-RS19 dataset. The coordinate attention achieves the suboptimal classification performance improvement. The CA and CBAM achieve the equivalent performance. Our SAA outperforms the other three attention mechanism lies in that the SAA infers attention maps along both channel and spatial dimensions, especially, it extracts mean values as well as affinities in each channel along two separate spatial directions as spatial information. This is different from that of CBAM, which uses a convolution operation with the filter size of $7 \times 7$ to explore local spatial information. The SAA can not only model long-range dependency, but also exploit global structural information, which prompts finer attention inferences on remote sensing images.

## V. Conclusion

In this article, we have proposed a novel SAA and CSSCL strategy to extract powerful and discriminative features for FSRSSC. The SAA first encodes spatial information of each channel along vertical and horizontal directions, respectively, which contains mean values and affinities. Then it infers attention maps according to the encoded spatial information. It can guide the network to focus on critical regions along both channel and spatial dimensions. Moreover, the proposed CSSCL strategy selects the corresponding class surrogate instead of the samples from the same class for the anchor samples as positives and the samples from the different classes as well as the remaining surrogates as negatives. It enhances intra-class compactness and inter-class dispersion by pulling positives together and pushing the negatives apart. The experimental results on three public remote sensing scene classification datasets demonstrate the effectiveness of our proposed method.

In our work, a single feature vector for each input images is extracted by pooling the feature maps, which may lead to information loss. In our future work, we will focus on using local descriptors to represent the remote sensing images. And the image relations based on local descriptors will be explored to improve the FSRSSC performance.

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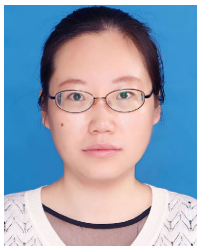
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